Group 4
Nguyễn Ngọc Thảo An – 103240210
Lê Hồ Thu Giang – 106240409
Nguyễn Đặng Quốc Đạt – 106240045

KAPPA FOUNDATION

Initial Decision Record | Case decision: 31 December 2022

Proposed policy. Reserve sufficient default-free case-bond principal for the two fixed liabilities; invest the remaining initial capital in a constrained growth portfolio. The 2030 real-value objective is a target, not a guaranteed outcome. We select Sample Mean-Variance (MV) for growth, conditional on the stated assumptions and risk review rules.

1 Candidate universe

Candidate Equity Universe

10 US common stocks/ 6 sectors
2 US - listed equity ETF candidates

No.	Security type	Sector/ ETF mandate	Ticker
1	US Common Stock	Technology	AAPL
2	US Common Stock	Technology	MSFT
3	US Common Stock	Financials	JPM
4	US Common Stock	Financials	BAC
5	US Common Stock	Health care	JNJ
6	US Common Stock	Health care	PFE
7	US Common Stock	Consumer staples	PG
8	US Common Stock	Consumer staples	KO
9	US Common Stock	Energy	XOM
10	US Common Stock	Industrials	CAT
11	Equity ETF	US large-cap value	VTV
12	Equity ETF	US small-cap	IWM

These offer distinct mandates; a zero portfolio weight is permitted. Eligibility as at the case date must be documented in the dated data record.

2 Estimation window

Estimation Window

Parameter	Specification
Estimation Period	January 2015 – December 2022
Data Frequency	Monthly
Data Source	yfinance
Asset Coverage	12 equity candidates + VTI benchmark
Return Measurement	Adjusted-price return proxies
Sample Size	Target: 96 common monthly returns, subject to validation
Annualisation	Mean return × 12; Sample covariance × 12

Data treatment: December 2014 adjusted prices are required to calculate January 2015 returns. Only observations available through the 31 December 2022 information cutoff are used for portfolio estimation and selection. January 2023 - December 2025 is reserved for out-of-sample evaluation.

3 Benchmark

We select the Vanguard Total Stock Market ETF (VTI) as a diversified, investable benchmark for the growth portfolio. VTI represents the broad US equity market and provides a simple alternative against which to evaluate our investment strategy. For a fair comparison, both portfolios use the same entry date, transaction costs, dividend treatment, and evaluation period. The bond allocation is assessed separately based on contractual cash flows and interest-rate scenarios.

4 Initial dollar allocation (USD 12 million)

Asset / purpose	Initial cost (USD)	Scheduled coverage
2,400 Beta zero-coupon bonds	$2,214,672.00	$2.4m principal, 31 Dec 2024
3,600 Delta 3.5% bonds	$3,549,852.00	$3.6m principal, 31 Dec 2027
Growth equities (net purchases)	$6,229,246.75	Eight selected stocks
One-time equity entry cost (10 bp)	$6,229.25	Charged at inception
Initial operating cash buffer	$0.00	No liquidity reserve at inception
TOTAL	**$12,000,000.00**	**All initial cash allocated**

Growth purchases (share of growth / USD): Eight positive holdings; BAC, XOM, VTV and IWM receive zero weight.

Ticker	Weight	Purchase (USD)	Ticker	Weight	Purchase (USD)
AAPL	15.00%	$934,387.01	PG	15.00%	$934,387.01
MSFT	15.00%	$934,387.01	KO	12.59%	$784,421.14
JPM	4.24%	$264,259.39	CAT	14.90%	$927,950.62
JNJ	9.10%	$566,729.50	PFE	14.17%	$882,725.07

Funding cash flows: Beta pays $2.4m in 2024. Delta pays $126,000 coupons each year 2023-2027 and $3.6m principal in 2027. Base policy holds the $630,000 accumulated coupons in cash at 0% through 2030; coupons are neither added to initial growth capital nor counted twice against the liabilities.

5 Material assumptions and decision evidence

Investment constraints. We follow the case restrictions on borrowing, short selling and derivatives. Each direct stock is limited to 15% of the growth portfolio, with a 30% limit per direct-stock sector to control concentration risk.

Return and risk assumptions. Expected annual returns combine 25% of each asset's historical mean and 75% of VTI's historical mean, annualised by multiplying by 12. This reduces reliance on individual stock estimates. We use A = 3 for the base utility comparison and A = 6 for sensitivity analysis. The utility measure covers the equity sleeve, not the entire fund.

Inflation and funding. We assume 2.3% annual inflation, using the 30 December 2022 US 10-year breakeven rate as a planning reference rather than a forecast. The initial cash buffer is $0, while Delta's coupons are retained at 0% interest. We also test a $250,000 buffer. Equity entry costs are set at 10 bps, including for VTI, with dividends reflected in adjusted-price return proxies rather than counted twice.

Bond valuation. We derive the first three spot rates from Alpha, Beta and Gamma. Since the fourth-year rate is unavailable, we assume s4 = s3 and infer the fifth-year rate from Delta. We test ±100 bps parallel rate shifts and changes in curve shape to assess valuation risk.

Our decision. Under the base assumptions, Sample Mean-Variance provides the highest estimated equity utility among the tested portfolios. Minimum-Variance remains a lower-risk alternative. The preference between these two fixed portfolios reverses at approximately A = 8.91. We therefore recommend Sample MV under our stated assumptions, while recognising that the result is sensitive to risk preferences and return estimates.

Growth strategy	Expected annual return	Annual volatility	Equity utility (A=3)
Sample MV - selected	12.05%	14.98%	0.0535
Single-index MV	11.91%	14.46%	0.0533
Minimum-Variance - rejected	10.86%	13.15%	0.0494

Risks and trade-offs. A -30% equity shock plus a +100 bp yield shift for Beta and Delta produces an estimated initial whole-fund mark-to-market decline of about 17.22% using exact bond repricing in the workbook. This breaches the trustees' area of concern near 15%; it is not a return forecast or acceptable-loss guarantee. Matched maturities still meet the two contractual payments if held to maturity under the case's default-free assumption. At 2.3% inflation, the 2030 real-value goal implies roughly $9.596m nominal; with coupons held at 0%, growth equities require about 4.66% annual geometric growth, not a promised return.

6 Evaluation and rebalancing rule (frozen policy)

Implementation: Establish the fixed dollar allocations above. For evaluation, use fractional equity purchases at the 3 January 2023 closing-price proxy after 10 bp entry costs, and apply the same rule to VTI. This execution convention is specified before viewing the subsequent performance period.

Rebalancing: No discretionary trading and no resetting equity weights to targets during 2023-2025. Allow weights to drift; model cash dividends reinvested in the same security via adjusted-close total-return proxies. Handle mandatory corporate actions consistently, with no voluntary tender or exchange elections.

Monitoring: Report the growth sleeve's compound return, volatility, maximum drawdown and active return relative to frozen VTI over 2023-2025. Evaluate bonds through contractual cash flows, duration/convexity and explicit yield scenarios rather than fictitious observed bond prices. Record liabilities as withdrawals, not investment losses.

Review triggers: A modelled, withdrawal-adjusted trailing 12-month whole-fund loss of 12% prompts an early warning; at 15% the investment committee must review the policy within 10 business days. A review does not itself authorise retrospective rebalancing. Any post-22 October change, including the trustees' revised mandate, must be logged against this original policy with its date and evidence.